



One song in. Every stem out.

SoundView v2 is a music-first stem separator. Import or record any audio, separate it on-device into as many stems as the model can find — vocals, drums, bass, guitar, keys, anything — then mix, solo, trim, and export each lane. Same Pine/Fern/Amber identity, same scroll-the-wave physics, now multiplied across stacked lanes.

01 Vision & Flow

02 Screens

03 Stem lane anatomy

04 Architecture

05 Components

06 Testing

01 — VISION & USER FLOW

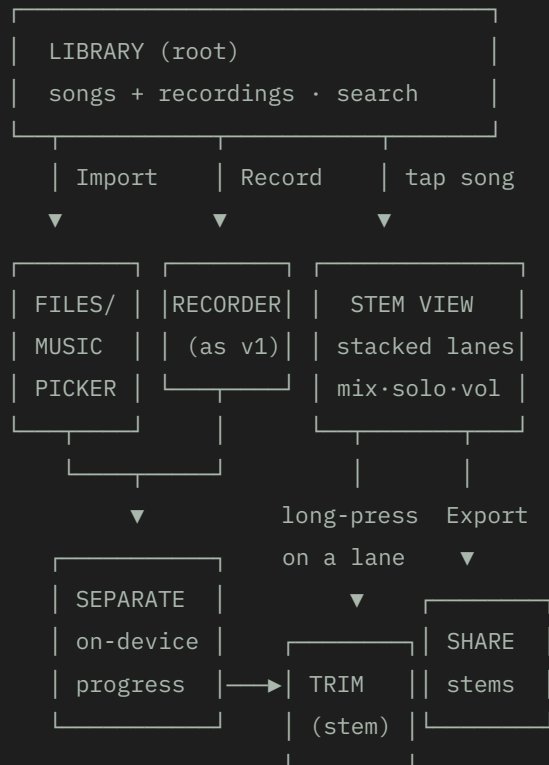
The song is a folder. The stems are its files.

Files-first, still. Each song becomes a human-named package in the library: the original plus its stems. Users see “Midnight Drive · 5 stems” — never a folder path. Everything syncs through iCloud as before.

Unbounded stems. No fixed 4-stem grid. The separation model emits whatever it confidently finds — 2 stems or 12 — and the UI is a vertically scrolling stack of lanes that handles any count.

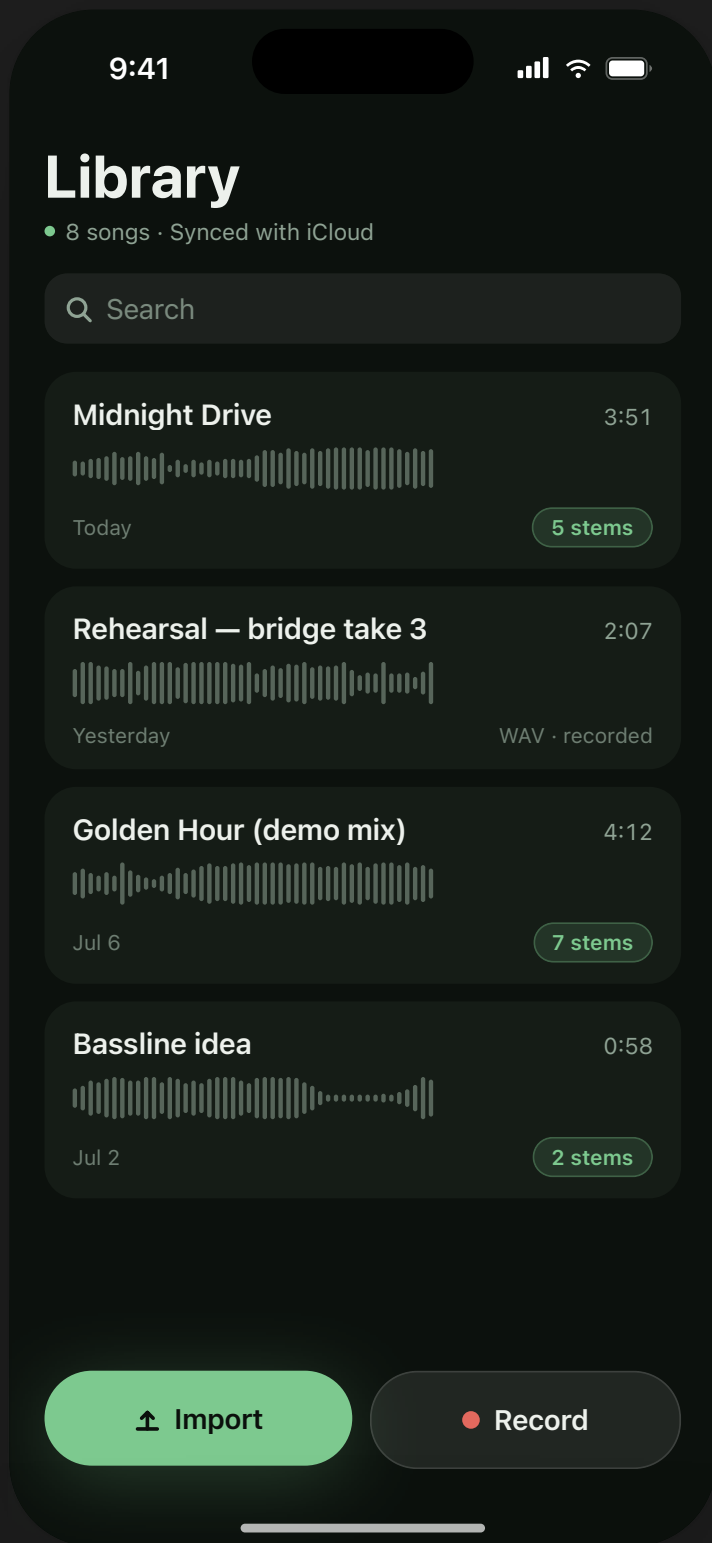
Two inputs, one pipeline. Import (Files / music library) or Record — both land in the library, both can be separated. Separation runs on-device (Core ML), offline, with honest progress.

Trim recedes. Trim/crop survives but per-stem and tucked behind a long-press — the main verbs are now *separate*, *mix*, *export*.



Library → Separate → Stem View.

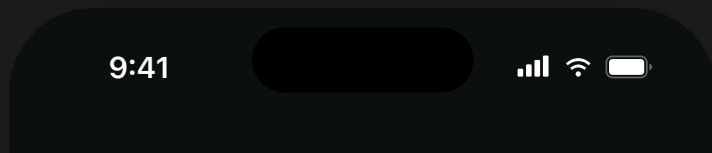
The Recorder is unchanged from v1 §2.2. Tweaks: wave seed, playback position, stem count.



2.1 Library

Same bones as v1, two changes: the bottom bar now holds **Import** (primary — the main way music enters) and **Record** (secondary), and separated songs carry a Fern **stem-count badge** (“5 stems”). Unseparated items show their format instead.

- **Import sources** — Files, music library, share-sheet (“Open in SoundView”), drag-and-drop on iPad/Mac. WAV, M4A/AAC in; MP3 open-only, as aligned.
- **Tap behavior** — separated song → Stem View. Unseparated → Sound View (v1 single-wave) with one prominent “Separate” action.
- **Human storage** — a song and its stems are one package; delete the song, stems go with it. UI says “Midnight Drive and its 5 stems” — never shows the folder.
- **A11y** — row reads “Midnight Drive, 3 minutes 51, separated into 5 stems, synced.”



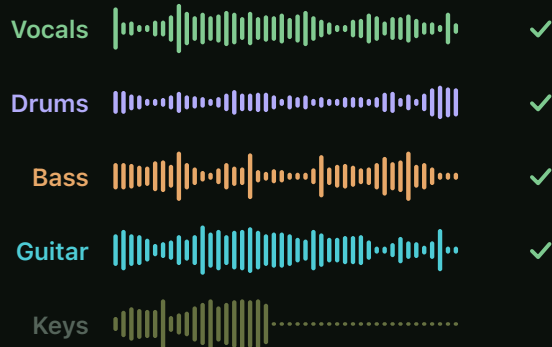
2.2 Separation

On-device Core ML is slow — so the wait is the show.

● SEPARATING

Midnight Drive

On this iPhone · works offline



Found 4 stems so far...

~1 min left

You can leave this screen — separation continues in the background and we'll notify you.

Cancel

Stems **materialize one by one** as the model finds them: each new lane fades in with its waveform drawing left-to-right. Finished stems get a Fern check; pending ones sit dim. No fixed stem list — the screen grows to whatever the model emits.

- **Honest progress** — count found so far + rough time remaining. Never a fake spinner.
- **Backgroundable** — leave anytime; processing continues (audio background task), local notification on completion; the library row shows a progress ring meanwhile.
- **Early listening** — tapping a finished stem's wave auditions it immediately, before the rest complete.
- **A11y** — VoiceOver announces each stem as found: "Drums separated. 4 of about 6."

2.3 Stem View — the new heart

Stacked lanes, one per stem, any count — the stack scrolls vertically past ~6 lanes. The playhead needle spans **all lanes at screen center** and never moves;

horizontal drag on any lane scrolls *all* waves together (they're one timeline). Pinch zooms all lanes in sync. Same physics as v1, multiplied.

- **Per-lane controls** — volume slider, Mute, Solo in each lane header. Muted lanes dim to 35%; soloed lane gets an Amber SOLO tag while others gray out. Multiple solos allowed.

9:41

Midnight Drive

5 stems · 3:51

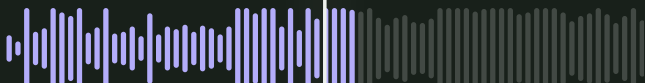
Vocals



M

S

Drums



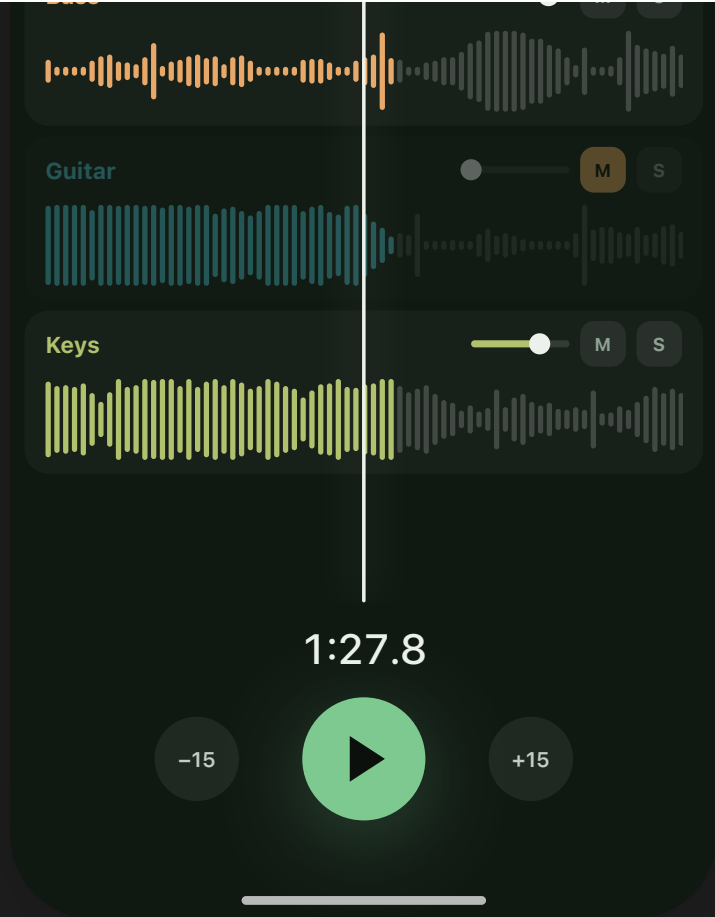
M

S

Bass

M

S

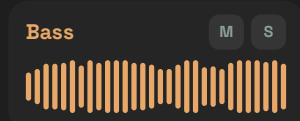


- **Lane colors** — each stem gets a hue derived from the palette (Fern, Amber, and oklch siblings — teal, moss, clay...) assigned in order; deterministic per song.
- **Long-press a lane** — context menu: *Trim stem* (v1 trim UI scoped to that lane — the "hidden" trim), *Export stem*, *Rename*, *Reset volume*.
- **Export** — share icon top-right: "Current mix" (respects mute/volume), "All stems" (zip of files), or pick stems. WAV or M4A out.
- **Align** — each lane is a container: "Bass, volume 80%, playing. Actions: mute, solo, trim, export." Needle position adjustable via rotor, announced as timecode.

03 — STEM LANE ANATOMY & STATES

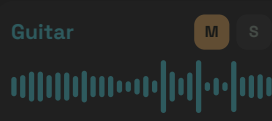
One lane, four states.

DEFAULT



Audible, mixed. Tap wave to seek all lanes; drag horizontally to scrub the shared timeline.

MUTED



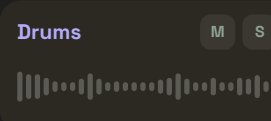
Lane dims to 35%, M fills Amber. Still scrubs and scrolls — only silent.

SOLOED



S fills Fern, lane bg tints; all non-soloed lanes gray out. Multiple solos combine.

TRIM (long-press)

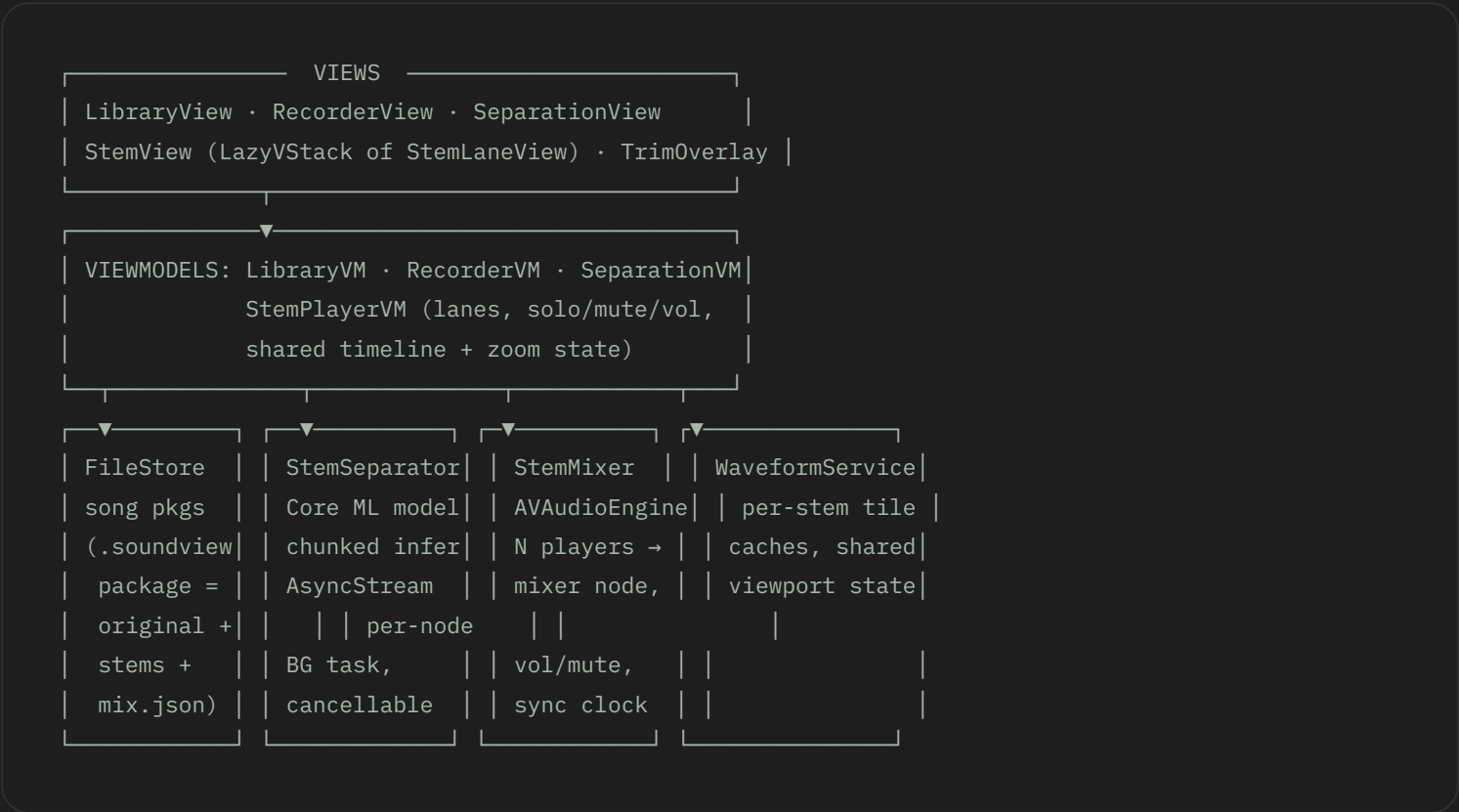


Lane expands to full height with v1 Amber trim handles; other lanes collapse. Save writes only this stem.

Palette rule for unlimited stems. Lane hues are generated as `oklch(0.78 0.11 h)` stepping hue from Fern's 150° in golden-angle increments, skipping the red band (reserved for record state). Two stems or fifty, every lane stays distinguishable, harmonious with Pine, and AA-contrasted. Trim mode inside a lane reuses the v1 Amber handles unchanged — same component, narrower canvas.

v1 architecture + two services.

Everything from v1 S05 stands — MVVM, @Observable, protocol services, files as truth. The pivot adds StemSeparator and StemMixer, and upgrades FileStore to song packages.



Song package.

A document package (shows as one item in Files): original audio + one file per stem + `mix.json` (volumes, mutes, lane names/colors, model version). Syncs atomically via iCloud; the whole thing is portable.

Separation pipeline.

Audio → chunked windows → Core ML source-separation model → per-source masks → reconstructed stems written incrementally. Emits `AsyncStream<StemEvent>` (.found, .progress, .finished) so the UI materializes lanes live. Model is swappable — stem count is whatever it outputs.

Synchronized playback.

One AVAudioEngine, N AVAudioPlayerNodes into a mixer, sample-locked start times so lanes never drift. Volume/mute are node properties — zero-latency, no re-render. Export “current mix” = offline render of the same graph.

v1 components + five new.

COMPONENT	RESPONSIBILITY	USED IN
-----------	----------------	---------

COMPONENT	RESPONSIBILITY	USED IN
StemLaneView	One lane: header (name, volume, M/S) + WaveformView bound to the shared viewport. Handles long-press menu.	Stem View
StemStackView	LazyVStack of lanes + the single cross-lane playhead overlay; owns vertical scroll.	Stem View
SharedViewport	Observable timeline state (position, zoom, velocity) every lane reads — one drag moves all waves.	Stem View, Trim
SeparationProgressView	Materializing stem list driven by AsyncStream<StemEvent>; per-stem check, progress bar, ETA.	Separation, Library row ring
StemBadge	Fern capsule "N stems" count.	Library rows
ExportSheet	Current mix / all stems / pick stems; format choice; offline render progress.	Stem View share

WaveformView, TrimHandles, TransportBar, TimecodeLabel, CapsuleButton, SyncStatusLine, EmptyState carry over from v1
\$06 unchanged; folder structure gains `Features/Stems/` and `Services/StemSeparator.swift` · `StemMixer.swift`.

06 — TESTING & EDGE CASES DELTA

New failure surface: the model and the mixer.

UNIT Mix state mix.json round-trip; volume/mute/solo matrix resolution (solo overrides mute rules); lane color determinism per song.	INTEGRATION Separation lifecycle Cancel mid-run (partial stems cleaned up); app killed during separation (resume or safe restart); low-battery/thermal throttle; model missing → graceful download prompt.	INTEGRATION Sync drift N players sample-locked over a full song; drift after seek, ±15s skips, route change mid-play; mute/unmute glitch-free.
EDGE Model output 1-stem result (mono speech), 12+ stems, silent stems (offer to discard), very long songs (chunk memory ceiling, suggest trimming first).	EDGE Package integrity Stem file missing from package (re-derive or mark broken lane); iCloud conflict on mix.json (last- writer-wins, audio never lost); import DRM-protected file → clear refusal message.	PERF Many lanes 60fps scroll with 10 lanes × 1-hour audio (lazy lane rendering); separation ≤ ~1× realtime on A16+; export render off main thread with progress.

Next step. Review the three new screens and the lane anatomy. Open questions worth deciding together: default export format (WAV vs M4A), whether "Separate" should auto-run on import or stay manual, and max song length for on-device processing. v1 remains the reference for Recorder, design tokens, and base architecture.